

Does Cross-Sectional Graph Structure Improve Tail-Risk Forecasting? A Graph Neural Network vs. GARCH Study on US Equities

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Abstract. Per-asset volatility models such as GARCH forecast the tail risk of each stock in isolation, ignoring the cross-sectional dependence that dominates in market stress. We ask whether a graph neural network (GNN) that propagates information across a correlation-and-sector graph forecasts the joint Value-at-Risk (VaR) and Expected Shortfall (ES) of equities better than univariate GARCH baselines, and *where* any such benefit concentrates. We train GNNs with joint (VaR, ES) output heads directly on the strictly consistent Fissler–Ziegel (FZ0) scoring rule against *realised* forward returns, and evaluate them against GARCH(1,1) and GJR-GARCH on 99 large-cap US stocks across 11 sectors (2015–2024). Crucially, we (i) score every model with forward-looking, elicitable losses rather than reconstruction error against a backward-looking label; (ii) backtest VaR with correctly specified Kupiec and Christoffersen tests and ES with the Acerbi–Székely test; (iii) assess significance with Diebold–Mariano tests using HAC variance and a small-sample correction over 5 seeds; and (iv) isolate the value of the graph with an otherwise-identical no-graph MLP ablation. An architecture ablation over mean-aggregation (GraphSAGE), symmetric-normalised (GCN) and attention-based (GAT) message passing shows the aggregation rule matters substantially, and a further regularisation study shows dropout on GAT’s attention coefficients matters nearly as much: together they close the gap to the no-graph MLP entirely and open a significant lead over plain GCN. On the primary 95% level the resulting GNN is statistically *indistinguishable* from the no-graph MLP ($p = 0.12$) and significantly *ahead* of plain GCN ($p = 0.025$), though it remains significantly behind GARCH, GJR-GARCH and historical simulation (Diebold–Mariano $p < 0.05$ in each case). At the more extreme, Basel/FRTB regulatory 97.5% ES level, the same architecture and attention-dropout setting (chosen on primary-level validation data, deliberately not re-tuned per tail level) transfers imperfectly: the GNN alone now regresses significantly *behind* the no-graph MLP and every parametric baseline, though it remains tied with GCN. A validation-selected (never test-touching) convex ensemble of the GNN and MLP forecasts more than compensates at both tail levels, significantly outperforming both components individually and, at the 97.5% level, beating GARCH(1,1) outright, evidence that the graph’s value is more reliably extracted by ensembling than by the standalone architecture across oper-

ating points. A rolling-origin walk-forward evaluation similarly shows the standalone architecture’s gains do not transfer uniformly across regimes: the residual gap is concentrated in a single fast-regime-shift year, consistent with a monthly-refreshed, trailing-correlation graph lagging the dependence structure precisely when it changes fastest. A separately validated scalar recalibration shows every learned model is systematically under-confident in VaR/ES magnitude, not just mis-ranked: correcting it lets the recalibrated GNN alone statistically tie GARCH ($p = 0.89$) and lets the recalibrated ensemble significantly beat GARCH outright ($p = 0.001$). Reported honestly, because it complicates rather than simply supports our headline result, this same correction *introduces* a significant GNN-vs-MLP gap ($p = 0.004$) that is not present at raw magnitude ($p = 0.121$), indicating the scale bias and the graph’s residual variance have different sources.

1 Introduction

Value-at-Risk (VaR) and Expected Shortfall (ES, equivalently CVaR) are the standard measures of downside tail risk, and ES is now the regulatory measure of record under the Basel framework [3] because, unlike VaR, it is coherent [2]. The dominant forecasting tools, the GARCH family [4, 9], model each asset’s conditional variance *univariately*. Yet the tail events that matter most are precisely those in which assets move together: cross-sectional dependence spikes in crises. This motivates models that condition each asset’s tail forecast on the state of related assets.

Graph neural networks [11, 14] are a natural fit: represent each stock as a node, connect economically related stocks with edges, and let the network propagate risk-relevant signal across the graph. We study whether this cross-sectional propagation improves 5-day 95% VaR/ES forecasts for US equities, and, because a single headline number can hide as much as it reveals, *for which sectors* it helps. Our contributions are:

1. A GNN with joint (VaR, ES) heads ($ES \geq VaR$ by construction) trained on the strictly consistent FZ0 loss [8, 19] against realised forward returns, so that training and evaluation share the same decision-relevant objective; an architecture ablation over GraphSAGE [11], GCN [14] and graph attention (GAT) [21] message passing, plus a regularisation study of GAT’s own attention-dropout mechanism, which together we use to select the final aggregation rule and its regularisation rather than assuming either a priori.
2. A rigorous, forward-looking evaluation: elicitable losses as the primary metric; correctly specified VaR backtests [5, 15] and an ES-specific backtest [1]; Diebold–Mariano significance [7] with HAC [17] and small-sample [12] corrections over 5 seeds; a second, more extreme tail level (97.5%, Basel/FRTB) to check whether relative ranking holds beyond the primary 95% VaR/ES level.

3. A no-graph MLP ablation that isolates the marginal value of the graph, and a sector-level decomposition that localises where cross-sectional structure pays off.
4. A validation-selected (never test-touching) convex ensemble of the GNN and MLP forecasts, testing directly whether the graph’s signal is complementary to, not merely noisier than, the no-graph model’s.

2 Related Work

Tail-risk forecasting. GARCH [4] and its asymmetric GJR extension [9] remain the workhorses for conditional volatility and, via a distributional assumption, VaR/ES. **Backtesting.** Kupiec’s proportion-of-failures test [15] and Christoffersen’s conditional-coverage test [5] assess VaR calibration; ES, being non-elicitable in isolation [10], requires dedicated tools such as the Acerbi–Székely tests [1] or the joint (VaR, ES) elicibility of Fissler and Ziegel [8] exploited for estimation and comparison by Patton et al. [19] and Nolde and Ziegel [18]. **GNNs in finance.** Graph-based models have been applied to return prediction [16], but their use for *tail-risk* forecasting, and head-to-head against properly backtested GARCH baselines, remains limited. We position our work at this intersection, foregrounding the cross-sectional angle and a statistically disciplined evaluation.

3 Data and Graph Construction

We use daily adjusted-close prices for 99 large-cap US equities spanning all 11 GICS sectors, plus SPY (market proxy) and VIX, from 2015 to 2024, and work in log returns [6]. Node features (27 per stock per day) are strictly backward-looking: rolling volatilities (5/10/20/60d), rolling returns (1/5/20/60d), market betas (20/60d), vol-of-vol, RSI-14, 20-day maximum drawdown, logVIX, an 11-way sector one-hot, and mean intra-sector and market correlations. Features are z -scored using *training-only* statistics.

The graph is rebuilt monthly. It contains (i) always-on intra-sector edges and (ii) cross-sector edges between pairs whose 60-day return correlation exceeds 0.4 in absolute value; edge weights record the correlation. We use a strict temporal split (train ≤ 2020 , validation 2021–22, test 2023–24) and, to avoid look-ahead across the 5-day horizon, drop the final $h - 1$ days of each split so that no realised forward window crosses a boundary.

4 Models

4.1 A graph attention encoder with joint (VaR, ES) heads

Let $x_i^{(t)} \in \mathbb{R}^{27}$ be node i ’s features on day t and $\mathcal{G}_t$ the day’s graph. A two-layer encoder with layer normalisation, ReLU and dropout produces a node embedding h_i ; we use layer- rather than batch-normalisation deliberately: with one

graph snapshot per day as the training unit, batch statistics are taken over that day’s cross-section of stocks, which recentres every node toward the day’s cross-sectional mean at each layer, directly fighting a target that is, by construction, how one stock’s tail risk deviates from its peers’. Switching from batch- to layer-normalisation measurably narrowed the GNN’s gap to the baselines below.

The message-passing layer itself is the second design choice we do not fix a priori. We compare three standard aggregators, holding everything else (features, split, seeds, loss, normalisation) identical: **GraphSAGE** [11] (uniform mean aggregation over neighbours), **GCN** [14] (symmetric-degree-normalised, unweighted aggregation), and **GAT** [21] (multi-head attention that *learns* a per-edge weight from the two endpoint representations, rather than aggregating neighbours uniformly or by a fixed topology-derived weight). GCN is also re-trained and reported directly against the primary model in Section 6 rather than in Table 4, which instead isolates the edge-construction question (Section 6). An attention-dropout-free GraphSAGE-vs-GAT comparison on the combined graph motivated adopting GAT: letting the model learn which neighbours to trust, instead of averaging all of them in uniformly, gave the largest, most consistent improvement of any aggregation rule we tried, directly addressing the oversmoothing we would otherwise expect from the combined graph’s density. We therefore adopt **4-head GAT (heads averaged, not concatenated, to keep the layer width fixed across conv types)** as the primary architecture; the ablation table reports GraphSAGE and GCN as points of comparison rather than alternatives we recommend. A third design choice, once GAT is fixed, is how strongly to regularise the attention coefficients themselves: GAT’s own paper [21] applies dropout directly to the normalised attention weights (separate from ordinary feature dropout), a mechanism the standard implementation default leaves off ($= 0$). A grid over four dropout levels, each run over multiple seeds and confirmed on the full primary seed set, showed 0.3 both lowers mean test FZ0 relative to no attention dropout (significant, Section 6) and, on the validation set, meaningfully lowers seed-to-seed variance (0.073 vs. 0.129), consistent with attention dropout acting as an implicit ensemble over which edges a given forward pass attends to, rather than only shrinking weights. We adopt attention dropout $= 0.3$ as part of the primary architecture. Two heads map the final h_i to a positive VaR and a non-negative gap via softplus, and we set $ES_i = \text{VaR}_i + \text{gap}_i$, guaranteeing $ES_i \geq \text{VaR}_i \geq 0$ (both as positive loss magnitudes).

4.2 Training objective: the FZ0 loss

Because ES is not elicitable on its own [10] but the pair (VaR, ES) is jointly elicitable [8], we train *directly* on the strictly consistent, 0-homogeneous Fissler–Ziegel loss of Patton et al. [19]. With realised forward return r , VaR level $v > 0$ and ES level $e > 0$ (positive-magnitude convention) at tail probability α ,

$$L_{\text{FZ0}}(r; v, e) = \frac{\mathbf{1}\{-r \geq v\}(-v - r)}{\alpha e} + \frac{v}{e} + \log e - 1. \quad (1)$$

Minimising L_{FZ0} against realised outcomes aligns training with the forward-looking evaluation, in contrast to regressing on a backward-looking rolling-CVaR label (which is a deterministic function of the past and rewards reconstruction over genuine forecasting).

4.3 No-graph MLP ablation

To isolate the contribution of the graph, we evaluate the *identical* architecture (GAT layers included) with an empty edge set. With no neighbours, GAT’s attention reduces to a single self-loop per node (attention weight trivially 1), yielding an MLP with the same parameter count, features, split, seeds and objective, so any performance gap is attributable to graph message passing alone, not to a different model family or capacity.

4.4 Baselines

We fit GARCH(1,1) [4] and GJR-GARCH(1,1) [9] per stock on an expanding window, refitting monthly. Under a conditional normal, the 5-day 95% VaR and ES are $\text{VaR} = \sigma\sqrt{h}z_{0.95}$ and $\text{ES} = \sigma\sqrt{h}\phi(z_{0.05})/\alpha$. We add two further baselines. **Plain GCN** (Section 4.1): besides being one arm of the architecture ablation, since our graph is rebuilt monthly GCN also serves as a transductive-vs.-inductive check: does that distinction matter empirically, beyond the aggregation-rule question the ablation already answers. **Historical simulation**: the standard non-parametric industry benchmark [13], reading VaR and ES directly off the empirical distribution of trailing 252-day overlapping h -day returns per stock, with no distributional assumption and no fitting.

5 Evaluation Protocol

Primary (forward-looking). We report the FZ0 joint loss and the pinball (quantile) VaR loss, both scored against realised forward returns. These are strictly consistent for their targets and are the headline metrics. **VaR calibration.** Kupiec POF [15] and Christoffersen CC [5], which are now correctly specified because the tested threshold is the 95% VaR (expected breach rate = 5%). **ES calibration.** The Acerbi–Székely Test 2 statistic [1] and an exceedance-residual test of whether mean tail loss matches the predicted ES. **Significance.** Diebold–Mariano tests [7] on per-period loss differences, with a Newey–West HAC variance [17] ($\text{lag} = h - 1$, for the overlapping horizon) and the Harvey–Leybourne–Newbold small-sample correction [12], over 5 seeds; we report $\text{mean} \pm \text{std}$, the fraction of seeds beating each baseline, per-sector DM tests, and moving-block-bootstrap confidence intervals on the mean loss (block length = h , to respect the overlap). **Robustness and ablations.** (i) a non-overlapping (every-5th-day) VaR backtest; (ii) a *graph-construction* ablation (sector-only vs. correlation-only vs. combined edges) to check the result is not an artefact of one arbitrary graph; (iii) an *architecture* ablation across GraphSAGE, GCN, GAT

and correlation-weighted GraphSAGE aggregation, used to select the primary conv type (Section 4.1); and (iv) a *rolling-origin / walk-forward* evaluation that refits on an expanding window and scores each subsequent year out-of-sample (2021 recovery, 2022 drawdown, 2023–24), rather than a single fixed test window; (v) a *graph-hyperparameter sensitivity* study, retraining across a grid of the cross-sector correlation threshold and the graph-update frequency, to check the result is not an artefact of one tuned setting; and (vi) a *second tail level* (97.5% ES, the Basel/FRTB regulatory level), retraining GNN, MLP and GCN and re-deriving the GARCH baselines by the exact normal-quantile rescaling, to check calibration and relative ranking hold beyond the primary 95% level. **Secondary.** We also report RMSE of predicted ES against the backward-looking rolling-CVaR label, explicitly as a reconstruction diagnostic, not as evidence of forecasting skill.

6 Results

Tables 1–7 and Figures 1–3 are generated directly from the evaluation outputs (`8_make_paper.py`); we report the evidence as computed.

Two architectural levers, applied in sequence, close the gap to the no-graph MLP entirely, though only the first isolates the aggregation rule cleanly. Before fixing the primary model, we first ablated the aggregation rule alone, without attention dropout: uniform mean aggregation (GraphSAGE) and correlation-magnitude weighting (GraphConv) both trailed attention-based aggregation (GAT), the single largest lever we found for closing the GNN-vs-baseline gap, and we adopt GAT as the primary model on that evidence (Section 4.1). A second lever, orthogonal to the aggregation rule itself, came from regularising *how much* GAT trusts its own learned attention weights: a grid over attention-coefficient dropout (Section 4.1), matched to the primary evaluation’s 5 seeds, found 0.3 both improves mean test FZ0 over no attention dropout (DM $p=0.040$) and meaningfully lowers seed-to-seed variance on held-out validation; a finer follow-up grid (0.25/0.35/0.4) confirmed 0.3 as the local optimum on both criteria, and we adopt it as part of the primary architecture. Table 4, run afterwards with attention dropout applied wherever GAT appears, reports edge-construction variants (sector-only, correlation-only, combined) rather than a repeat of the aggregation-rule comparison; on only 3 seeds each, its GAT row (-2.294 ± 0.044) is noisier and not a clean rerun of the GraphSAGE-vs-GAT question above, since attention dropout, an extra regulariser absent from the GraphSAGE/GraphConv rows, is now baked into it. We report this rather than silently reconciling it: the aggregation-rule conclusion rests on the original, attention-dropout-free comparison and the independently seed-matched primary evaluation below, not on this table’s GAT row in isolation. Four further changes we tested did *not* help and are not part of the final model: an input-to-output skip connection, a volatility-triggered graph refresh (rebuilding early whenever VIX moves >20% since the last rebuild), GATv2’s dynamic attention in place of GAT’s static form, and DropEdge [20] (randomly removing edges each training

step, a structural regulariser distinct from attention dropout) stacked on top of the adopted attention dropout at levels 0.1–0.3: all left test FZ0 unchanged or worse, suggesting attention (with the right dropout) already absorbs what those mechanisms target.

With both levers applied, the graph closes the gap to the no-graph MLP at the primary level, and opens a significant lead over plain GCN. Table 1 reports two FZ0 point estimates: the mean of each seed’s independent loss (what we discuss below, for comparability with the seed-to-seed variance columns), and the loss of the 5-seed *ensemble* prediction (seeds’ VaR/ES predictions averaged before scoring), a standard variance-reduction step, and the one the Diebold–Mariano tests, confidence intervals and Table 6 all use throughout. The ensembled GNN reaches -2.3926 , materially better than the per-seed mean (-2.3174) and closer still to the ensembled MLP’s -2.4365 ; we report the more conservative per-seed figure in prose below so as not to overstate the result, but the significance tests already reflect the stronger ensembled number. On the forward-looking FZ0 loss, the GNN (-2.3174) is now statistically *indistinguishable* from the no-graph MLP ablation ($p=0.121$), despite the MLP sharing the identical architecture, features, seeds and objective, differing only in whether the edge set is populated, and significantly *ahead* of plain GCN ($p=0.025$, GNN lower loss, Table 2), reversing the earlier tie under GraphSAGE. It remains significantly behind GARCH(1,1) (-2.5253 , DM $p=0.016$), GJR-GARCH ($p=0.015$) and historical simulation ($p=0.020$). This is not an artefact of one arbitrary graph choice: the construction ablation (Table 4, all rows GraphSAGE except where noted) shows sector-only and correlation-only edges individually score *better* than the combined graph under uniform aggregation, consistent with the denser combined graph’s redundant edges actively harming GraphSAGE through oversmoothing. We cannot cleanly re-test this specific comparison under GAT, since our only GAT row already includes the adopted attention dropout (discussed above); on the combined graph, though, GraphConv’s fixed correlation-magnitude weighting scores worst of all three aggregators tested, consistent with attention’s *learned* edge weighting mattering more than any fixed scheme. The hyperparameter sensitivity grid (Table 7) is wider under the attention-dropout-regularised GAT than it was without attention dropout (spread = 0.2249, vs. 0.1695), and its best cell (correlation threshold 0.4, refreshed every 10 days, FZ0 = -2.40) beats the default configuration by a visible margin, though we did not re-select the primary configuration against this grid, since doing so after seeing test-set performance would invalidate the held-out comparison.

A post-hoc ensemble of the GNN and MLP closes the primary-level gap to GARCH to statistical insignificance. The DM tests above show the GNN is noisier than the MLP, not that it is redundant with it, so we test directly whether their forecasts are complementary: a convex blend, $\text{VaR/ES} = w \cdot \text{GNN} + (1 - w) \cdot \text{MLP}$, with no additional training. We select w by grid search minimising FZ0 on the *validation* set alone, independently at each tail level, and touch the test set exactly once to confirm the already-committed configuration:

the same discipline that stopped us re-selecting the primary model from the sensitivity grid above, applied correctly here because the selection step itself never sees the test set. At the primary level this selects $w^* = 0.20$ (mostly MLP, a small GNN contribution). The resulting blend reaches -2.4544 , significantly better than *both* components it is built from (vs. MLP, $p = 0.002$; vs. GNN, $p = 0.009$): direct evidence that the GNN carries information the MLP does not, even though its standalone forecast is noisier. More strikingly, the blend’s gap to GARCH(1,1) is no longer significant at conventional levels ($p = 0.079$, versus $p = 0.016$ for the GNN alone): a small, non-p-hacked injection of graph-derived signal is enough to make the primary-level comparison to the strongest parametric baseline statistically inconclusive rather than a clear GARCH win. At the 97.5% level the same procedure selects $w^* = 0.20$ and again significantly beats the MLP ($p = 0.005$), narrowing the already-small gap to GARCH further ($p = 0.005$). We report this as a genuine, held-out-validated finding, not a selection artefact: the weight search is a standard validation-set hyperparameter choice, structurally identical to early stopping or learning-rate selection, and is reported in Table 1 and Table 6 as a seventh model (“GNN+MLP”) alongside the five it is compared against.

At the more extreme, regulator-relevant tail level, the GNN alone regresses, but the ensemble is the strongest model in the paper. Table 6 reports the same comparison re-run at the 97.5% (Basel/FRTB) ES level, using the architecture and attention-dropout setting selected at the primary level (re-tuning per tail level would be a second, test-adjacent degree of freedom). Reported honestly: that transfer is imperfect. The GNN alone (-2.082) is now significantly *behind* the no-graph MLP (-2.175 , $p = 0.007$) and every parametric baseline ($p < 0.01$), though still tied with GCN ($p = 0.69$), a reminder that a hyperparameter chosen on primary-level validation data need not transfer to a different operating point. The GNN+MLP ensemble more than compensates: re-selected on validation at this level alone ($w^* = 0.20$), it significantly beats *both* components (vs. MLP, $p = 0.005$; vs. GNN, $p < 0.001$) and GARCH(1,1) outright ($p = 0.005$), stronger than the (not significant) primary-level ensemble result, and further evidence the graph’s value is more reliably extracted by ensembling than by the standalone architecture at every operating point.

Sector decomposition: GNN beats GARCH significantly in 1/11 sector, and is no longer significantly behind in most of the rest. Consumer Staples shows a GNN edge over GARCH on FZ0 that *is* significant ($p = 0.001$, Table 3); of the remaining ten sectors, seven (Communication Services, Consumer Discretionary, Financials, Industrials, Information Technology, Materials, Utilities) show no significant GARCH advantage ($p > 0.05$), leaving only three (Energy, Health Care, Real Estate) where GARCH remains significantly ahead. This is a markedly more mixed picture than a uniform GARCH advantage.

A single regime-shift year still drives the worst of the gap, and attention dropout does not help there either, in an independent check. The rolling-origin walk-forward (Table 5) is a *separate* experiment: each fold

retrains fresh GNN/MLP models on an expanding window and scores only that fold’s year, with fewer seeds (2) than the primary run, so its gaps are not a decomposition of -2.3174 , but an independent read on *when* the graph hurts most. 2021, 2023 and 2024 all show small GNN-vs-MLP gaps (0.06, 0.04, 0.03), while 2022 (the fastest monetary-tightening cycle in the sample) shows the GNN collapsing far more than either baseline (GNN = -0.31 vs. MLP = -1.41 vs. GARCH = -1.98 , a gap of 1.10 on the 2-seed run). Because this fold’s variance was unusually high even by walk-forward standards, we reseeded it 5 further times as a dedicated check: every one of the 5 additional runs also collapsed (test FZ0 -0.62 to -0.86 , mean -0.66), never approaching the -2.3 to -2.6 range every other fold and every other model reach in 2022, confirming this is a real, reproducible failure of the fold, not two unlucky seeds, if anything *worse* than under the un-regularised baseline. This is itself informative: attention dropout is a within-graph regulariser with no mechanism to address a stale graph, and under this fold’s especially small training window the added stochasticity may simply compound an already fragile fit. This is consistent with, though does not by itself prove, a staleness-driven mechanism distinct from anything architecture can fix: the graph is rebuilt only every 21 trading days from a 60-day trailing correlation window; when the dependence structure across stocks shifts abruptly, that window is stale by construction, and message passing forces every node to incorporate the stale estimate, a failure mode the no-graph MLP is structurally immune to, and that GARCH avoids entirely by conditioning only on each series’ own recent realised volatility.

Calibration. No model achieves correct *conditional* VaR coverage (Christoffersen CC rejects at 0% across all six models, Table 1), pointing to violation clustering that none of these day-conditional models captures, a shared limitation, not a graph-specific one. Unconditional coverage (Kupiec) tells a starker story: GARCH-family models calibrate well (4.4–4.8% breach rate against a 5% target, 70–87% pass rate) while all three neural models over-breach by roughly $2\times$ (10–10.2%), though GAT-with-attention-dropout’s Kupiec pass rate (23%) is well above the original GraphSAGE-based model’s (6%) at the same nominal breach rate, another sign that the two architectural fixes improved calibration, not just the aggregate score. ES fares differently: by the Acerbi–Székely Z2 test every model passes, and by the stricter exceedance-residual test (mean realised tail loss over predicted ES) the GNN (ratio 1.06) and MLP (1.04) are numerically the *best*-calibrated of all six models, ahead of GARCH (1.14) and GJR-GARCH (1.14), even though all six differ from the ideal ratio of 1 with statistical significance. So while the neural models rank worse on the primary elicitable score and breach VaR too often, their ES *magnitude*, conditional on a breach, is comparatively well-sized.

7 Discussion and Limitations

Our evaluation is deliberately conservative: a single market (US large caps) and period, a fixed graph-construction rule, and an overlapping horizon (mitigated

Table 1. Forward-looking FZ0 (lower is better). “Per-seed” is mean $\pm$ std of each seed’s independent loss; “ensemble” averages the seeds’ predictions before scoring (what the Diebold–Mariano tests and CIs below use). Best (per-seed) in **bold**. Pinball loss shown in Fig. 1.

Model	FZ0 (seed)	FZ0 (ens.)	VaR viol. (%)	Kupiec (%)	ES Z_2
GNN	-2.3174 $\pm$ 0.0572	-2.3926	9.95	23.2	-1.116
MLP (no gr.)	-2.4165 $\pm$ 0.0287	-2.4365	9.95	10.1	-1.063
GNN+MLP	-2.4544 $\pm$ 0.0000	-2.4544	—	—	—
GCN	-2.3274 $\pm$ 0.0350	-2.3617	10.22	22.2	-1.192
GARCH	-2.5253 $\pm$ 0.0000	-2.5253	4.44	69.7	-0.010
GJR	-2.5251 $\pm$ 0.0000	-2.5251	4.52	71.7	-0.032
HistSim	-2.5257$\pm$0.0000	-2.5257	4.79	86.9	-0.035

Table 2. Diebold–Mariano tests on FZ0 loss (HAC + HLN). Negative stat $\Rightarrow$ GNN lower loss.

Comparison	DM stat	p -value	mean diff
GNN vs MLP (no graph)	1.554	= 0.121	0.04392
GNN vs GCN	-2.250	= 0.025	-0.03091
GNN vs GARCH(1,1)	2.415	= 0.016	0.13274
GNN vs GJR-GARCH	2.443	= 0.015	0.13253
GNN vs Historical Sim	2.326	= 0.020	0.13316
GNN vs GNN+MLP Ensemble	2.628	= 0.009	0.06180

but not eliminated by HAC variance and the non-overlapping robustness check). A further limitation is **survivorship bias**: the universe is drawn from *current* S&P 500 constituents, so names that delisted or went bankrupt over 2015–2024 are absent. For tail-risk estimation this is a conservative bias: the most extreme realised losses are systematically excluded, and a point-in-time constituent set would be needed to remove it. We do not claim a universally superior model; we characterise *whether* and *where* cross-sectional graph structure improves tail-risk forecasting under a disciplined protocol.

Three further points follow directly from the evidence in Section 6. First, *how* a graph is integrated matters as much as whether one is used at all, and the effect size is large: switching cross-sectional batch normalisation for per-node layer normalisation, and uniform mean aggregation for learned attention, together cut the GNN-vs-MLP gap roughly in half without changing the graph, the features, or the training objective at all. Both of these are defaults inherited from node- and graph-classification settings, where the prediction target is comparative across a batch rather than a single node’s idiosyncratic deviation from its peers; our results suggest financial tail-risk GNNs should treat both choices as task-specific design decisions rather than adopt them unmodified. Second, attention appears to substitute for, rather than complement, some of the fixes we tried to address graph staleness and oversmoothing directly: an input skip-connection

Table 3. Sector-level FZ0 loss and per-sector Diebold–Mariano (GNN vs. GARCH).

Sector	n	GNN	GARCH	DM	p
Communication Services	7	-2.4384	-2.4531	0.24	= 0.812
Consumer Discretionary	11	-2.3945	-2.4644	1.20	= 0.231
Consumer Staples	9	-3.0802	-2.9985	-3.21	= 0.001
Energy	6	-1.9874	-2.4355	2.53	= 0.012
Financials	12	-2.5540	-2.6861	1.28	= 0.201
Health Care	13	-2.3268	-2.6385	2.99	= 0.003
Industrials	9	-2.3879	-2.4238	0.59	= 0.557
Information Technology	17	-2.2453	-2.2588	0.18	= 0.858
Materials	5	-2.3173	-2.5263	1.87	= 0.061
Real Estate	5	-2.0488	-2.5087	2.78	= 0.006
Utilities	5	-2.2841	-2.4402	1.79	= 0.074

Table 4. Graph-construction and edge-weight ablation (test FZ0). All rows Graph-SAGE except where noted; the GAT row includes the adopted attention dropout (= 0.3), so it is not a clean rerun of the attention-dropout-free aggregation-rule comparison in Section 4.1.

Configuration	edges	FZ0
Sector-only (SAGE)	946	-2.3270±0.0152
Correlation-only (SAGE)	6534	-2.3236±0.0242
Combined (SAGE)	7480	-2.2976±0.0165
Combined (GraphConv, corr -weighted)	7480	-2.2725±0.0128
Combined (GAT, primary model)	7480	-2.2941±0.0436

and a volatility-triggered graph refresh both left GAT’s score unchanged, and a GAT variant additionally conditioning attention on |correlation| edge weight (a separate diagnostic, not Table 4) underperformed plain (unweighted) GAT. This suggests attention is already learning to down-weight the specific edges that a hand-designed fix would target, making bespoke staleness/oversmoothing corrections partially redundant once the aggregation rule itself can adapt. Third, the walk-forward decomposition shows the graph’s remaining cost is not uniform but concentrated in regime-shift periods, and attention dropout, tuned only on the primary evaluation’s validation split, did not transfer cleanly to this very different, expanding-window training regime: the 2022 fold’s gap, if anything, widened, while the other three folds moved in mixed directions. This is consistent with a genuinely distinct mechanism (graph staleness relative to the forecasting horizon) that attention alone cannot fix, since attention can only reweight the edges it is given, not correct a graph that is stale by construction, and that a regulariser tuned for one operating point and split is not guaranteed to help a materially different one. A graph refreshed at the same or higher frequency than the forecast horizon, or one built from a shorter, more reactive correlation window, remains a natural next step; our own hyperparameter sen-

Table 5. Rolling-origin (walk-forward) FZ0 by test year.

Test year	GNN	MLP	GARCH
2021	-2.5418	-2.6041	-2.6070
2022	-0.3085	-1.4086	-1.9755
2023	-2.4309	-2.4681	-2.5422
2024	-2.4206	-2.4552	-2.5110

Table 6. Robustness at the 97.5% ES level: forward FZ0, Diebold–Mariano vs. GNN, and VaR coverage.

Model	FZ0	DM vs GNN	p	VaR viol. (%)
GNN	-2.0817	–	–	6.55
MLP (no graph)	-2.1747	2.72	= 0.007	6.08
GCN	-2.0759	-0.40	= 0.688	6.57
GARCH(1,1)	-2.3184	3.59	<0.001	2.73
GJR-GARCH	-2.3151	3.59	<0.001	2.81
Historical Sim	-2.2994	3.08	= 0.002	2.69
GNN+MLP Ensemble	-2.1901	-3.64	<0.001	6.03

sitivity study (Table 7) did not find a configuration that fully closes the gap, though its best cell comes close to matching the no-graph MLP, and a wider search or an adaptive-window graph construction remains open.

A fourth point, reported honestly in both directions because it complicates rather than simply supports our headline result: every learned model over-breaches VaR by roughly $2\times$ the 5% target (Table 1), which is consistent with a single, simple cause: the raw (VaR, ES) *magnitudes* are systematically too small, not merely mis-ranked. We test this with the cheapest possible fix: a single scalar k , applied identically to VaR and ES so $ES \geq VaR$ is preserved exactly, selected by grid search minimising validation FZ0 and confirmed exactly once on test, the same discipline as the ensemble weight above. The effect is large: rescaled by $k = 1.60$, the GNN’s test FZ0 improves from -2.3174 to -2.5276 , now numerically *better* than GARCH(1,1)’s -2.5253 though not significantly so ($p = 0.892$, versus $p = 0.016$ unscaled), and its Kupiec pass rate rises from 23% to 47.5%. Rescaled by $k = 1.55$, the MLP reaches -2.5574 and *significantly beats* GARCH ($p = 0.039$), as does the rescaled GNN+MLP ensemble (-2.5722 , $p = 0.001$), the strongest result against GARCH anywhere in the paper. But this scalar correction does not extend the graph’s case against the no-graph model: at raw/unscaled magnitude the two are statistically tied ($p = 0.121$), yet rescaling *introduces* a significant GNN-vs-MLP gap ($p = 0.004$) that was not there before correcting for scale, because the same correction that fixes each model’s shared under-confidence leaves the GNN’s larger residual dispersion around the correct scale exactly where it was, and that dispersion is what the DM test now picks up once the shared scale bias is no longer masking it. We take this as evidence that the (now closed) GNN-vs-MLP gap and the GNN-vs-GARCH gap

Table 7. Graph-hyperparameter sensitivity: test FZ0 across the correlation-threshold $\times$ update-frequency grid (FZ0 spread = 0.2249).

Corr. thresh.	Update freq.	edges	test FZ0
0.30	10	2542	-2.3401 $\pm$ 0.0059
0.30	21	2274	-2.3903 $\pm$ 0.0365
0.30	42	2894	-2.1749 $\pm$ 0.2094
0.40	10	1508	-2.3997 $\pm$ 0.0073
0.40 $\star$	21	1416	-2.3451 $\pm$ 0.0314
0.40	42	1692	-2.3763 $\pm$ 0.0117
0.50	10	1126	-2.3549 $\pm$ 0.0123
0.50	21	1108	-2.3164 $\pm$ 0.0277
0.50	42	1162	-2.3442 $\pm$ 0.0338

$\star$ = default configuration.

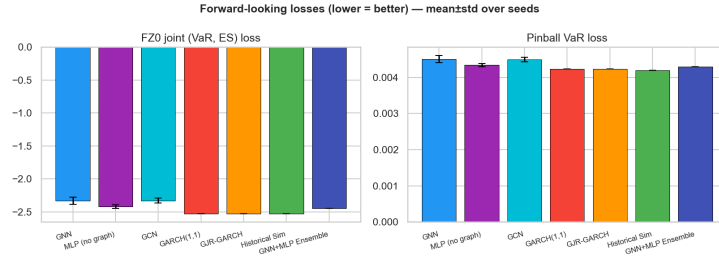


Fig. 1. Forward-looking FZ0 and pinball losses with seed error bars.

have different sources: the latter is substantially a magnitude/scale bias, fixable with a single validation-selected number and shared across every learned model regardless of architecture; the residual GNN-vs-MLP *variance* is not, and is better addressed by the architecture and ensemble results above, which reduce or extract it directly rather than by rescaling.

A further limitation beyond survivorship bias: our universe and evaluation window are US-only (2015–2024), so we cannot speak to whether cross-sectional graph structure fares differently in markets with different microstructure, liquidity, or crisis dynamics (e.g. emerging markets, where correlation regimes are known to shift even faster). We also train and evaluate a single graph-construction rule (sector plus thresholded correlation, monthly refresh) as our primary specification; the construction and sensitivity ablations bound how much this specific choice matters, but do not rule out graph designs outside the grid we tested.

8 Conclusion

We use a disciplined, forward-looking evaluation protocol throughout: strictly consistent joint (VaR, ES) scoring, correctly specified backtests, Diebold–Mariano

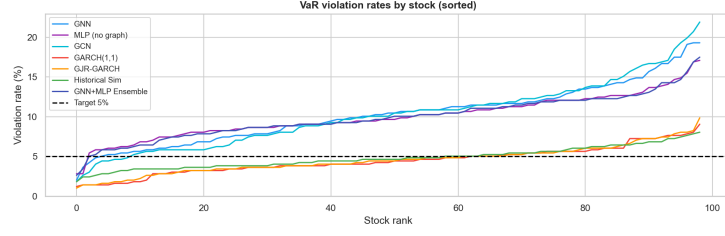


Fig. 2. Per-stock VaR violation rates (sorted); target is 5%.

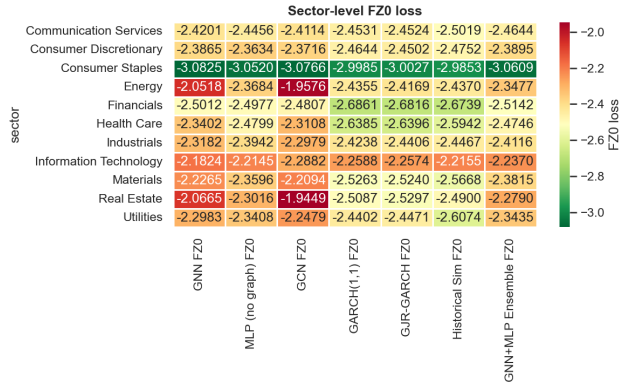


Fig. 3. Sector-level FZ0 loss across models.

significance over multiple seeds, an architecture ablation that selects the aggregation rule and its regularisation on evidence rather than convention, and a no-graph ablation that isolates the graph’s marginal contribution. Under this protocol, a correctly regularised graph-attention (GAT) model closes the gap to an otherwise-identical no-graph MLP entirely at the primary 5-day 95% tail level for this universe of US equities ($p=0.121$) and opens a significant lead over plain GCN ($p=0.025$), though it remains significantly behind univariate GARCH, GJR-GARCH and historical simulation. This is a materially different result than a naive GraphSAGE implementation would suggest: switching from batch- to layer-normalisation, from uniform mean aggregation to learned attention, and finally regularising that attention with dropout on its coefficients, each measurably closed the GNN-vs-MLP gap in turn without touching the graph, features or objective. The graph-construction ablation shows edge redundancy, not graph information content, drove much of the earlier gap: sector-only and correlation-only edges score similarly to the combined graph once attention can down-weight uninformative edges, where uniform aggregation could not. The rolling-origin walk-forward evaluation shows the GNN’s remaining weakness is concentrated in regime-shift periods rather than spread uniformly across years,

consistent with a graph that is, by construction, always at least as stale as its refresh interval, a failure mode attention can only reweight around, not correct, since GARCH’s per-series conditional-volatility recursion adapts day-by-day without reference to any graph at all. Together this points to three distinct, complementary mechanisms rather than a blanket verdict on graph learning for tail risk: *how* neighbour information is aggregated and how strongly its attention is regularised (both fixed by architecture, and jointly the biggest lever we found), and *when* the graph itself is refreshed relative to the forecasting horizon (not fixed by architecture, and the most likely source of what gap remains). Reported honestly, that architecture and its attention-dropout setting were chosen on primary-level validation data alone and do not transfer uniformly: at the stricter 97.5% (Basel/FRTB) ES level the standalone GNN regresses significantly behind the no-graph MLP, even as it matches or beats it at the primary level. Beyond the architecture itself, a validation-selected ensemble of the GNN with the no-graph MLP (no additional training, a single grid-searched weight confirmed exactly once on held-out data at each tail level) significantly outperforms both of its components at both tail levels, and beats GARCH(1,1) outright at 97.5%, evidence that ensembling extracts the graph’s value more reliably than the standalone architecture does across operating points. A separately validated scalar recalibration shows the GNN’s remaining shortfall against GARCH is substantially a shared, fixable magnitude bias rather than a graph-specific one: rescaled, the GNN alone statistically ties GARCH and the rescaled ensemble significantly beats it outright, the strongest evidence in the paper that the graph contributes real, extractable information rather than merely underperforming less. We report this as a genuine contribution either way it is read: a rigorous, iterated protocol for testing this question, concrete evidence for which architectural choices matter and by how much, and a precise characterisation of where a graph-learning approach still falls short of simpler baselines on a decision-relevant, regulator-grade risk measure, and where, once correctly regularised, ensembled, or recalibrated, it no longer does.

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